INVERTIS UNIVERSITY, BAREILLY

FACULTY OF ENGINEERING & TECHNOLOGY

OFFICIAL PRACTICAL & LABORATORY MANUAL â¢ DEPARTMENT OF CSE

Laboratory Course: [BCS-251] Programming for Problem Solving Lab

Programme: Bachelor of Technology in Computer Science & Engineering **Manual Type: LAB MANUAL**

Prescribed w.e.f Academic Session 2023-2024 **Lab Coordinator: Department of Computer Sc**

Official Laboratory Manual: C Programming & Problem Solving

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1. Course Objectives & Lab Evaluation

- Target: Foundational algorithmic problem-solving using C programming language.
- Assessment: Continuous lab assessment (20 Marks) + External Practical Examination (30 Marks).
- Compiler: GCC on Linux / Dev-C++ / CodeBlocks.

2. Prescribed Experiment Schedule

- Week 1-4: GCC setup, data types, arithmetic expressions, if-else branching, loops.
- Week 5-8: 1D arrays (searching & sorting), 2D matrix multiplication, custom functions.
- Week 9-12: Pointers, call-by-reference, string manipulation without string.h, structures.
- Week 13-14: Dynamic memory allocation (malloc/free) and persistent file input/output.